
Same Top Fraction, Different Workload: Auditing Eligibility and Capacity in Clinical Risk Evaluation

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Abstract

Clinical risk scores turn patient rankings into finite worklists for specialist review, monitoring, or prevention. Comparing the same top fraction across candidate sets of different sizes changes both who can be ranked and how many people receive service. We introduce a decision-aware four-cell audit that keeps the full-population event denominator fixed and crosses two eligibility rules with two absolute budgets, separating broader reach from additional capacity. We apply it to frozen birth-record score vectors for preterm birth, neonatal intensive care admission, and low birth weight in approximately 3.48 million CDC Natality 2023 records. At the nominal top 10% operating point, moving from early-entry eligibility to a retrospective all-record upper bound reports a 39–42% relative increase in captured events while expanding the worklist by one third. Symmetric accounting allocates about three fifths of the joint gain to added capacity and two fifths to broader eligibility; at a fixed budget, the eligibility gain is less than half the cohort-relative estimate. A complete replication on CDC 2024 preserves this allocation pattern. In 18 of 81 controlled settings, the mean cohort-relative contrast is positive while fixed-budget capture declines. The audit therefore distinguishes population reach from resource expansion and makes the policy represented by a top-fraction metric explicit. Code and reproduction materials are available at <https://github.com/vache4ogan/prenatal-risk-scoreability>.

1 Introduction

Clinical risk models increasingly determine which records receive specialist review, monitoring, outreach, or preventive services [Annarumma et al., 2019, Baltruschat et al., 2021, Jung et al., 2021, Singh et al., 2023]. These workflows are capacity constrained: a model produces a ranking, but an organization acts on a finite number of cases. Top- q precision, recall, or event capture therefore appears to connect predictive performance directly to an operational decision. Prenatal case-management studies, for example, evaluate the highest-risk 5–15% because program capacity covers a fraction of eligible pregnancies [Pan et al., 2017].

The concrete scale of this policy difference is visible in CDC Natality 2023 [National Center for Health Statistics, 2026]. An early-entry candidate set of approximately 2.60 million records yields a top-10% worklist of about 260,000, whereas an all-record set of approximately 3.48 million yields about 348,000 selections at the same nominal fraction. The comparison therefore changes both the candidate gate and the absolute workload. Clinical decision frameworks distinguish predictive

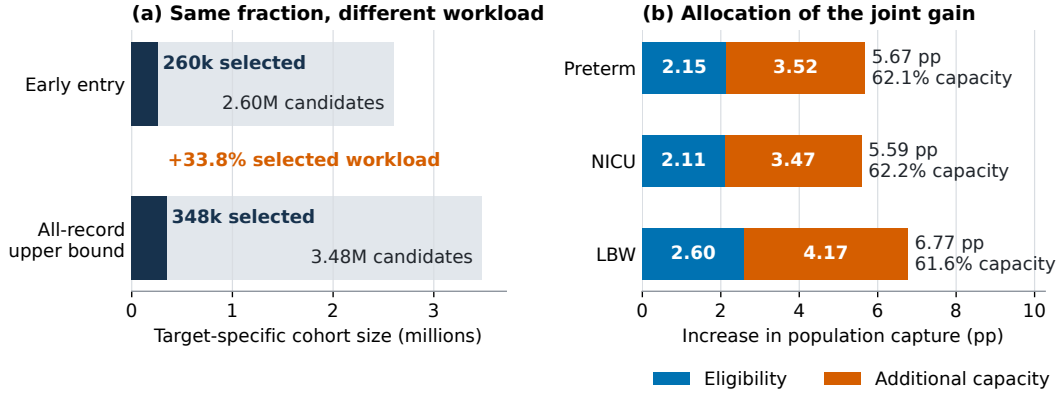


Figure 1: The same nominal top 10% encodes a different workload. Panel (a) summarizes the target-specific CDC 2023 cohorts: moving from the early-entry gate to the all-record upper bound increases the selected count from about 260,000 to 348,000. Panel (b) allocates the associated increase in population event capture at $q = 10\%$. Values inside bars are percentage-point allocations; capacity contributes about 62% of each total gain.

ranking from action under finite resources [Jung et al., 2021, Singh et al., 2023]: a fixed-capacity service and a service that expands with candidate volume implement different policies and require different contrasts.

Prenatal risk triage makes this ambiguity visible. Pregnancies enter care at different times, and adverse outcomes are concentrated among records with no documented prenatal care or unknown start timing. An early-entry score ranks events inside its candidate gate; events outside the gate contribute zero reachable cases regardless of discrimination. National vital statistics retain both the eligible and excluded outcomes, making it possible to distinguish retrieval within the gate from reach across the complete target population.

We audit preterm birth, neonatal intensive care unit (NICU) admission, and low birth weight in CDC Natality. Figure 1 previews how the workload change alters the interpretation of the reported gain.

Our contributions are:

1. **An explicit eligibility–capacity comparison design.** A shared score vector and population event denominator connect a same-fraction claim to four explicit policies. The two fixed-budget contrasts reveal which part survives equal workload; their interaction measures dependence on the chosen budget.
2. **A national-scale measurement study.** Frozen 2022 scoring pipelines applied to CDC 2023 show that the same top 10% expands workload by 33.8–33.9% and that capacity receives 61.6–62.2% of the symmetric allocation across three outcomes. A complete CDC 2024 replication tests temporal stability without refitting the scoring pipelines.
3. **A mechanism map and reporting profile.** An 81-condition stress test locates sign-reversal regimes, and the resulting checklist makes the candidate gate, absolute workload, and event denominator explicit.

2 Related Work

Risk ranking under limited resources. Prenatal case-management, imaging worklists, and clinical workflow studies connect risk scores to finite review or treatment capacity [Pan et al., 2017, Annarumma et al., 2019, Baltruschat et al., 2021, Jung et al., 2021, Singh et al., 2023]. These studies connect operating points to the capacity and value of a clinical workflow. Resource-constrained net benefit evaluates whether the expected value of acting on a score justifies its workload [Singh et al., 2023]. Calibrated subset selection constructs shortlists containing a desired number of quali-

fied candidates in expectation [Wang et al., 2022]. These objectives govern selection within a specified population and resource model. Interpreting a reported gain across candidate pools additionally requires tracking the absolute budget induced by each pool’s size.

Selective access to clinical data and services. Clinical records reflect who enters care, is tested, and receives a documented label. Informed presence and selection bias can alter both apparent risk and the population represented by an EHR cohort [Haneuse and Daniels, 2016, Goldstein et al., 2016, Sisk et al., 2021]; disparate censorship describes related variation in testing and label availability [Chang et al., 2022]. Health-service coverage frameworks separate service availability from realized coverage [Tanahashi, 1978]. This literature establishes that the candidate gate is informative, but a top-fraction comparison adds a second moving quantity because the number served scales with the observed cohort.

Evaluation validity and decomposition. Clinical reporting guidance requires explicit target populations, intended use, discrimination, and calibration [Collins et al., 2024]. Symmetric allocation across factor orderings is established decomposition machinery [Kitagawa, 1955, Shapley, 1953, Shorrocks, 2013]. The decomposition rule specifies how a joint difference is allocated once the comparison cells have been defined. Cross-cohort top-fraction evaluation still requires a choice of those cells: whose outcomes form the denominator, which records can enter the ranking, and how many positions each policy funds. Without these choices, a constant percentage can conceal a changing workload and leave the fixed-capacity conclusion unresolved.

3 Methodology

The audit fixes the target population, score vector, and event denominator, then crosses two candidate-set rules with two absolute budgets. Figure 2 shows the resulting four cells; each factor’s marginal change is averaged across the two orderings for symmetric allocation.

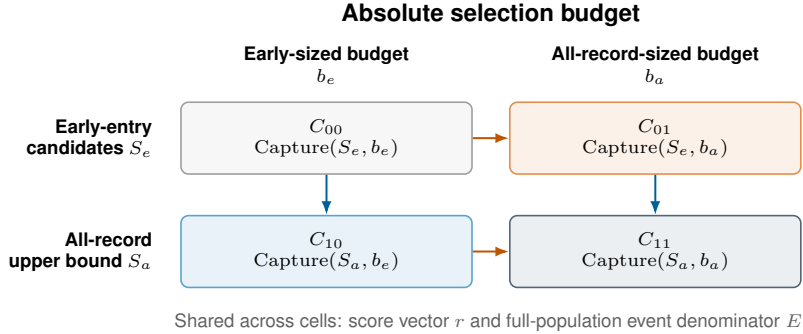


Figure 2: Eligibility–capacity audit. Rows vary the candidate set and columns vary the absolute selection budget while the score vector and full-population event denominator remain fixed. Blue arrows change candidate eligibility; orange arrows change capacity. For each factor, averaging its marginal capture change across the two orderings gives its symmetric Shapley allocation. The diagonal contrast $C_{11} - C_{00}$ is the usual same- q comparison and changes both factors.

3.1 Target population and scoring study

We use CDC Natality public-use live-birth records and official field definitions [National Center for Health Statistics, 2026, 2017]. CDC 2022 supplies model development with an 80/20 train/validation split; the frozen pipelines generate the shared score vectors analyzed in CDC 2023. Model fitting and hyperparameter selection use CDC 2022 exclusively; CDC 2023 is the temporal cohort in which the audit hypothesis was developed and evaluated. The target population contains U.S.-resident singleton live births with a known outcome: preterm birth below 37 completed obstetric-estimate weeks ($N = 3,480,382$, 303,001 events), NICU admission ($N = 3,476,584$, 303,285 events), and birth weight below 2,500 g ($N = 3,480,316$, 243,776 events).

Prenatal-care month defines four mutually exclusive groups: early entry in months 1–3, later documented entry in months 4–10, no prenatal care, and unknown start timing. The early candidate set contains months 1–3. The documented-start set contains months 1–10. The all-record set adds no-care and unknown-timing records and therefore bounds candidate and event availability under retrospective complete scoring. The care-month rule operationalizes documented entry into prenatal care rather than feature-level EHR availability; the birth-certificate model serves as a shared retrospective ranking instrument.

The baseline-oriented score uses nine maternal birth-certificate variables: race, height, pre-pregnancy BMI and weight, previous live births now living, previous live births now dead, other pregnancy outcomes, pre-pregnancy diabetes, and pre-pregnancy hypertension. Outcome, delivery, and neonatal fields are excluded. Numeric variables use development-set median imputation and standardization; race uses most-frequent imputation and one-hot encoding. One LightGBM model per outcome uses 350 trees, learning rate 0.05, 31 leaves, minimum child size 100, L2 penalty 1, and seed 2026 [Ke et al., 2017]. The audit compares candidate and budget policies on a shared score vector. A camera-ready replication applies the same frozen 2022 pipelines to CDC 2024, with identical cohort rules, four cells, and 500 paired resamples per outcome.

3.2 Decision metrics

Let $\mathcal{P}$ contain N records and $E = \sum_{i \in \mathcal{P}} Y_i$ events. For candidate set S , event count E_S , and integer budget b , deterministic tie breaking defines the b highest scores $\text{Top}(S, b)$. Their true-positive count is

$$\text{TP}(S, b) = \sum_{i \in \text{Top}(S, b)} Y_i. \quad (1)$$

We report both within-gate performance and population reach:

$$\text{Precision}(S, b) = \frac{\text{TP}(S, b)}{b}, \quad \text{Recall}_{\text{eligible}}(S, b) = \frac{\text{TP}(S, b)}{E_S}, \quad (2)$$

$$\text{Capture}(S, b) = \frac{\text{TP}(S, b)}{E}, \quad \text{Ceiling}(S) = \frac{E_S}{E}. \quad (3)$$

Population capture factors into retrieval and reach,

$$\text{Capture}(S, b) = \text{Recall}_{\text{eligible}}(S, b) \text{ Ceiling}(S). \quad (4)$$

Each result therefore reports target-population N and E , eligible n and E_S , exact selected n , precision, eligible recall, population capture, and the event ceiling. ROC-AUC and average precision (AP) provide a ranking-signal check; the four-cell contrasts carry the primary decision claim.

The eligibility ceiling concerns event availability before imposing a budget. With budget b , even perfect ranking is bounded by $\text{Capture}(S, b) \leq \min(E_S, b)/E$. The all-record candidate set therefore bounds reach, while its achieved capture remains a property of the ranking and budget.

3.3 Four-cell eligibility–capacity audit

Let $S_e \subseteq S_a$ be early and all-record candidate sets. A nominal fraction q induces $b_e = \text{round}(q|S_e|)$ and $b_a = \text{round}(q|S_a|)$. With $b_a \leq |S_e|$, the complete design is

$$C_{uv} = \text{Capture}(S_u, b_v), \quad u, v \in \{0, 1\}, \quad (5)$$

Here $S_0 = S_e$, $S_1 = S_a$, $b_0 = b_e$, and $b_1 = b_a$; event denominators are positive and $1 \leq b_e \leq b_a \leq |S_e|$. The usual cohort-relative contrast is $\Delta = C_{11} - C_{00}$. Its interaction is $I = C_{11} - C_{10} - C_{01} + C_{00}$. We retain both fixed-budget eligibility contrasts and summarize the joint change with the standard symmetric Shapley allocation [Shapley, 1953]:

$$\begin{aligned} \phi_E &= \frac{1}{2} [(C_{10} - C_{00}) + (C_{11} - C_{01})], \\ \phi_B &= \frac{1}{2} [(C_{01} - C_{00}) + (C_{11} - C_{10})]. \end{aligned} \quad (6)$$

The exact identity $\Delta = \phi_E + \phi_B$ holds, and $s_B = \phi_B/\Delta$ gives the symmetric capacity share for positive Δ . This signed allocation can exceed one when eligibility contributes negatively; it is undefined for a zero joint contrast. Symmetry assigns half the interaction to each factor rather than

identifying a causal share of the gain. For a frozen deterministic ranking, increasing b adds nested positions, so the capacity component is non-negative; expanding S at fixed b can have either sign. A uniformly random ranking supplies the mechanical reference $E[\text{Capture}(S, b)] = (b/|S|)(E_S/E)$.

The contribution is the comparison design: off-diagonal cells test equal-workload claims against a denominator retaining events outside the gate. The established Shapley rule summarizes these event-count changes; clinical effects require a separate intervention model.

For example, preterm capture at $q = 10\%$ is 14.63% in C_{00} , 17.89% in C_{01} , 16.52% in C_{10} , and 20.30% in C_{11} . Eligibility gains of 1.89 and 2.40 points at the two budgets average to $\phi_E = 2.15$ points; the 3.52-point capacity allocation completes the 5.67-point joint gain.

3.4 Evaluation and mechanism stress test

The empirical audit uses $q \in \{0.05, 0.10\}$. Point estimates are accompanied by 500 paired full-size empirical resamples per outcome and q . Each resample uses common record multiplicities across all four cells while holding the observed scores and absolute budgets fixed. The resulting central 95% ranges measure conditional stability of the reported four-cell contrasts.

The simulation draws latent risk $Z \sim \mathcal{N}(0, 1)$, outcomes with $P(Y = 1 \mid Z) = \text{logit}^{-1}(\alpha_Y + Z)$, eligibility with $P(A = 1 \mid Z) = \text{logit}^{-1}(\alpha_A + \gamma Z)$, and scores $R = Z + \sigma\epsilon$, where $\epsilon \sim \mathcal{N}(0, 1)$ is independent. Conditional on Z , outcome and eligibility draws are independent. Intercepts set marginal event prevalence to 0.10 and availability to a . Crossing $a \in \{0.5, 0.7, 0.9\}$, $\gamma \in \{-1, 0, 1\}$, $\sigma \in \{0.5, 1, 2\}$, and $q \in \{0.05, 0.10, 0.20\}$ gives 81 conditions. Each of 100 independent repetitions contains 100,000 evaluation records and 50,000 independent threshold-selection records, with shared random draws across conditions. Appendix C specifies all parameters, coupling, and generation steps.

4 Results

4.1 Candidate eligibility and event ceilings

Early entry contains about 74.7% of target-population records, but only 71.8% of preterm events, 70.6% of NICU events, and 69.4% of low-birth-weight events. Before ranking begins, the early gate therefore leaves 28.2–30.6% of events outside its reachable population. Including every documented prenatal-care start in months 1–10 raises the event ceiling to 91.1–92.1%.

The excluded records carry concentrated risk. Outcome prevalence among no-care records is 2.4–2.7 times the early-entry prevalence across the three targets; unknown-timing records are 1.7–1.8 times higher. Candidate expansion therefore changes both population reach and case mix, making a full-population event denominator essential.

Table 1: The candidate gate fixes reach before ranking. Early entry denotes prenatal-care initiation in months 1–3; documented start includes months 1–10. Risk ratios compare outcome prevalence in excluded groups with the early-entry group.

Outcome	Early eligible n	Early share	Early ceiling	Documented ceiling	No-care / unknown risk
Preterm birth	2,601,060	74.7%	71.8%	91.1%	$2.7\times / 1.8\times$
NICU admission	2,597,274	74.7%	70.6%	92.1%	$2.4\times / 1.7\times$
Low birth weight	2,600,093	74.7%	69.4%	91.6%	$2.6\times / 1.8\times$

4.2 Eligibility–capacity accounting under top- q selection

Figure 1 and Table 2 connect the nominal operating point to its actual workload. Across outcomes, the selected count increases from approximately 260,000 to 348,000. Population event capture rises from 14.63% to 20.30% for preterm birth, 13.93% to 19.52% for NICU, and 16.06% to 22.83% for low birth weight. These are relative gains of 38.8%, 40.1%, and 42.2%, respectively.

The proportional policy adds 87,931–88,023 service slots per outcome and captures 16,505–17,177 additional events. The yield of this combined policy change is 18.8–19.5 additional events per

100 added slots, equivalent to 5.1–5.3 additional selections per extra captured event. This ratio includes both expanded eligibility and the 33.8–33.9% larger capacity. Capacity-only yields compare columns within a fixed row of the four-cell design.

Table 2: Same nominal top 10%, different operational result. Stacked counts and capture report early entry above all records. ϕ_E and ϕ_B are the symmetric eligibility and capacity allocations. Conditional ranges appear in Appendix B, Table 8.

Outcome	Selected n	Workload	Capture (%)	Rel. gain	ϕ_E/ϕ_B (pp)	s_B
Preterm birth	260,106	+33.8%	14.63	+38.8%	2.15/ 3.52	62.1%
	348,038		20.30			
NICU admission	259,727	+33.9%	13.93	+40.1%	2.11/ 3.47	62.2%
	347,658		19.52			
Low birth weight	260,009	+33.9%	16.06	+42.2%	2.60/ 4.17	61.6%
	348,032		22.83			

The four-cell audit changes the interpretation of those gains. Symmetric accounting assigns 2.11–2.60 percentage points to eligibility and 3.47–4.17 points to additional capacity, placing 61.6–62.2% of the joint contrast on the workload increase. Holding the early budget fixed yields eligibility gains of 1.89–2.27 points; holding the all-record budget fixed yields 2.32–2.93 points. Both are materially smaller than the joint 5.59–6.77-point cohort-relative contrast.

The pattern persists at $q = 5\%$, where capacity receives 63.9–65.6% of the symmetric allocation. The smallest lower bound across the conditional ranges is approximately 60.4%. A random ranking allocates 79.7–87.9% to capacity at $q = 10\%$, showing that a majority capacity allocation can arise mechanically under a proportional budget. The learned ranking retrieves more of the added candidate groups’ event mass than the random reference, shifting the allocation toward eligibility. Complete four-cell counts and both operating points appear in Appendix B.

4.3 Fixed-budget and transported-threshold policies

Table 3 compares three policies on the same frozen score vector. Cohort-relative top-10% evaluation jointly changes eligibility and workload. Fixing the all-record absolute budget isolates candidate expansion: capture increases by 2.32–2.93 points, with exactly the same number selected on both sides. A threshold chosen once on the CDC 2022 validation split selects 9.15–9.82% of early-entry records and 10.20–10.40% of all records in CDC 2023. Under that transported threshold, the all-record worklist is 1.42–1.49 times larger and capture increases by 6.36–8.19 points.

Table 3: One score vector, three resource policies at the nominal 10% operating point. Counts and population capture compare early-entry with all-record eligibility. The score threshold is chosen on CDC 2022 and transported unchanged to CDC 2023.

Outcome	Resource policy	Selected n : early $\rightarrow$ all	Capture: early $\rightarrow$ all	Δ (pp)
Preterm birth	Cohort-relative top 10%	260,106 $\rightarrow$ 348,038	14.63 $\rightarrow$ 20.30%	+5.67
	Fixed absolute b_a	348,038 $\rightarrow$ 348,038	17.89 $\rightarrow$ 20.30%	+2.40
	Transported 2022 threshold	245,487 $\rightarrow$ 361,709	14.06 $\rightarrow$ 20.84%	+6.78
NICU admission	Cohort-relative top 10%	259,727 $\rightarrow$ 347,658	13.93 $\rightarrow$ 19.52%	+5.59
	Fixed absolute b_a	347,658 $\rightarrow$ 347,658	17.20 $\rightarrow$ 19.52%	+2.32
	Transported 2022 threshold	255,166 $\rightarrow$ 361,714	13.76 $\rightarrow$ 20.12%	+6.36
Low birth weight	Cohort-relative top 10%	260,009 $\rightarrow$ 348,032	16.06 $\rightarrow$ 22.83%	+6.77
	Fixed absolute b_a	348,032 $\rightarrow$ 348,032	19.90 $\rightarrow$ 22.83%	+2.93
	Transported 2022 threshold	237,973 $\rightarrow$ 355,034	14.98 $\rightarrow$ 23.17%	+8.19

4.4 Temporal replication

We apply the frozen CDC 2022 models and preprocessing to approximately 3.51 million CDC 2024 records per outcome, retaining the same candidate rules and four-cell estimands with a shared 2024

event denominator for each outcome. Table 4 reports both operating points with 500 paired resamples per outcome. At $q = 10\%$, the larger candidate set expands workload by 35.0–35.1% and increases captured events by 40.3–44.0% relative to early entry. Capacity receives 60.9–62.8% of this joint contrast, compared with 61.6–62.2% in CDC 2023. All conditional capacity-share ranges remain above 59.7%, and both fixed-budget eligibility contrasts remain positive for each outcome and budget. At the all-record-sized 10% budget, the eligibility gains are 2.32–3.06 points, compared with 5.66–7.01 points under the joint policy.

Table 4: CDC 2024 four-cell replication with the frozen 2022 pipelines. Capture changes and allocations are percentage points; capacity shares include central 95% ranges from 500 paired resamples.

Outcome	q	Joint gain	Eligibility	Capacity	Capacity share (%)
Preterm birth	5%	3.45	1.16	2.29	66.4 [64.9, 68.0]
Preterm birth	10%	5.85	2.23	3.62	61.9 [60.8, 63.0]
NICU admission	5%	3.34	1.15	2.19	65.6 [64.3, 67.0]
NICU admission	10%	5.66	2.11	3.55	62.8 [61.8, 63.9]
Low birth weight	5%	4.13	1.52	2.61	63.1 [61.7, 64.6]
Low birth weight	10%	7.01	2.74	4.27	60.9 [59.8, 61.9]

Transporting the full-precision 2022 cutoffs to CDC 2024 selects 10.60% of preterm, 10.64% of NICU, and 10.20% of low-birth-weight records: 21,112, 22,400, and 6,952 more slots than a 10% budget. Thus, freezing a threshold preserves a score cutoff while the realized workload can change with the year. The camera-ready replication corrects the race-category representation used in the reviewed 2024 threshold-only calculation; Appendix B documents the correction and unchanged CDC 2023 replay.

4.5 Controlled mechanism stress test

Figure 3 maps the protocol difference across all 81 synthetic conditions. Eighteen conditions report a positive cohort-relative gain while candidate expansion at the all-record absolute budget reduces population capture. Nine have a central replicate range entirely below zero. Every reversal arises when higher-risk records are preferentially available, so the newly admitted pool is lower risk and can displace selected events.

In Figure 3, at 70% availability, medium score quality, and $q = 10\%$, for example, cohort-relative capture increases 5.52 points while fixed-budget candidate expansion changes capture by -0.28 points; 93 of 100 repetitions reverse. The largest mean absolute interaction is 9.49 points. In CDC 2023, both fixed-budget eligibility contrasts remain positive for every outcome and q , so the ambiguity changes effect magnitude. The controlled grid shows that selective availability can also change effect direction.

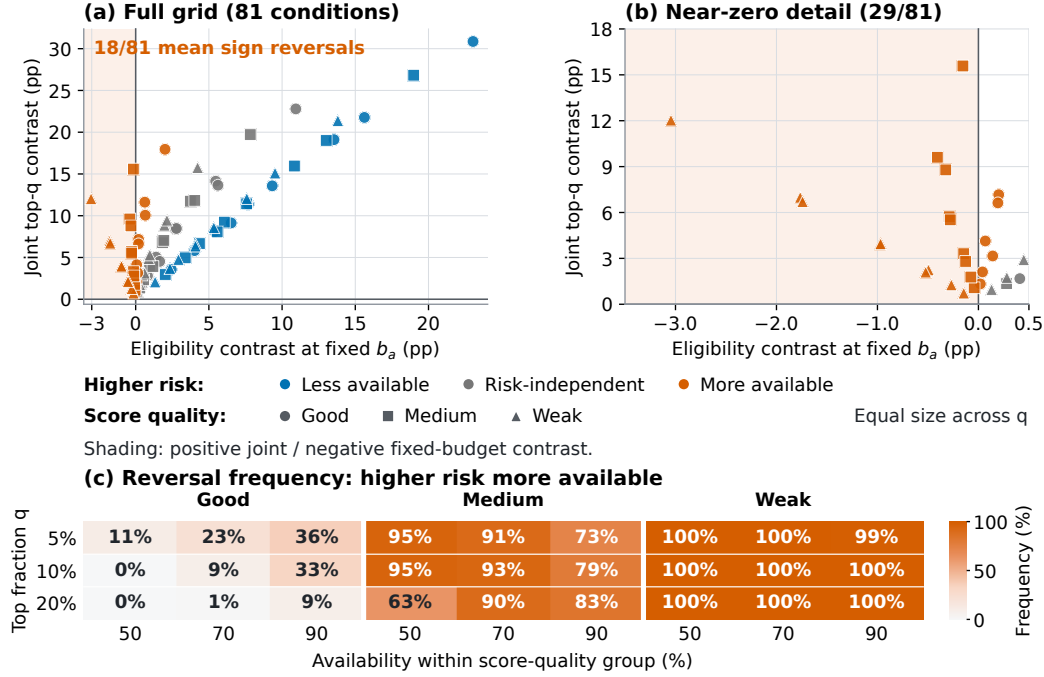


Figure 3: Full-grid mechanism stress test. Panel (a) compares the cohort-relative conclusion with candidate expansion at the all-record-derived absolute budget b_a . The shaded quadrant contains sign reversals: the joint top- q contrast is positive while fixed-budget eligibility expansion is negative. Panel (b) enlarges this region; each point is a condition mean over 100 repetitions. Panel (c) gives replicate-level reversal frequencies when higher risk is more available, grouped by score quality and eligible fraction. Marker shape denotes score quality.

4.6 Temporal score diagnostics

The frozen 2022 pipelines retain similar discrimination in CDC 2023 (Table 5). Across outcomes, ROC-AUC and average precision change by at most 0.003 from 2022 validation to 2023. Brier skill relative to a prevalence-only forecast is 1.7–2.4%. The four-cell audit fixes each transported score vector, isolating within-year policy contrasts from temporal score drift.

Table 5: Frozen-score temporal check. Brier baseline is the constant CDC 2023 prevalence prediction; skill is $1 - \text{Brier}_{\text{model}} / \text{Brier}_{\text{baseline}}$.

Outcome	ROC-AUC: 2022 $\rightarrow$ 2023	AP: 2022 $\rightarrow$ 2023	Brier: model / baseline	Brier skill
Preterm birth	0.614 $\rightarrow$ 0.612	0.146 $\rightarrow$ 0.146	0.0780 / 0.0795	1.9%
NICU admission	0.611 $\rightarrow$ 0.608	0.139 $\rightarrow$ 0.141	0.0783 / 0.0796	1.7%
Low birth weight	0.653 $\rightarrow$ 0.652	0.133 $\rightarrow$ 0.132	0.0636 / 0.0651	2.4%

4.7 Sensitivity and robustness

Two feature ablations preserve the allocation pattern. Removing maternal race and refitting each outcome model at $q = 10\%$ leaves capacity shares of 60.9–62.4%. A separate eight-feature specification removes prior other pregnancy outcomes (prior_terminations) and repeats the complete audit at both budgets with paired resampling. Capacity still accounts for 61.5–64.3% of the joint contrast. Interaction point estimates range from +0.32 to +0.69 percentage points; all conditional ranges exclude zero. The result therefore persists across two retrospective feature ablations. Establishing feature availability at an early clinical encounter requires separate timestamped records.

5 Discussion

Resource interpretation of top- q evaluation. Top- q evaluation is coherent when service volume scales with the candidate population; a fixed budget answers the constrained-capacity question. Here, a 38.8–42.2% capture gain includes 33.8–33.9% more workload. Making this policy visible helps prevent unplanned demand from redirecting scarce services.

From audit to policy choice. Compare feasible policies at a fixed selected count B . With equal event values and equal selection and access costs, maximize captured events. An illustrative retrieval utility generalizes this rule [Singh et al., 2023]:

$$U(S, b) = v \text{ TP}(S, b) - c b - K(S), \quad (7)$$

where $v > 0$ values a captured event, c costs one selection, and $K(S)$ costs access to the candidate pool. Prefer changes satisfying $v \Delta \text{ TP} > c \Delta b + \Delta K$; at fixed $b = B$, maximize $v \text{ TP} - K(S)$. For the joint CDC change with $\Delta K = 0$, the break-even cost-to-value ratio is 0.188–0.195. Translating retrieval into clinical utility additionally requires intervention benefits and harms.

Applicability across deterministic score vectors. Capacity monotonicity holds for any fixed ranking. Numerical allocations depend on case mix, ranking quality, eligibility, and budget: 61.6–62.2% characterizes these CDC 2023 tasks and LightGBM scores. Feature ablations test specification sensitivity; the controlled grid maps sign-reversal mechanisms.

A recommended reporting profile. Table 6 specifies the minimum reporting items. Implementation in existing evaluation pipelines uses stored scores, outcomes, and candidate masks; common resampling weights preserve pairing.

Table 6: Minimum reporting checklist for cross-cohort top- q evaluation. The final column illustrates each item with the CDC 2023 audit.

Required item	Minimum report	Example from this study
Target population	Population rule, N , total events E , and event denominator	CDC 2023 singleton births; $N \approx 3.48\text{M}$; all target events remain in the denominator
Candidate gate	Eligibility rule, eligible n , E_S , and event ceiling E_S/E	Care entry in months 1–3; $n \approx 2.60\text{M}$; ceiling 69.4–71.8%
Resource rule	Fraction q or absolute budget b , with exact selected n	Top 10%; selected n : 260k→348k (+33.8–33.9% workload)
Operating result	Precision, eligible recall, and population event capture	Capture: 13.93–16.06%→19.52–22.83%
Cross-cohort audit	Four cells, fixed-budget contrasts, interaction, and symmetric allocation	Capacity receives 61.6–62.2% of the joint allocation

Scope and next validation. Prospective EHR landmarks would validate input and intervention timing beyond the retrospective care-entry proxy. CDC 2024 supports temporal stability within this source; additional model families and clinical settings would test broader transport. Heterogeneous benefits, acquisition costs, and subgroup priorities require explicit utility and distributional analyses: removing race tests score specification, rather than establishing equitable access. Retraining for an expanded candidate set likewise defines a different comparison from this audit of fixed rankings.

6 Conclusion

Top-fraction comparisons encode service decisions as well as ranking quality. In CDC 2023, the same top 10% expands workload by one third, with about 62% of the capture gain allocated to capacity; the CDC 2024 replication preserves this pattern. Controlled experiments also identify opposite-sign conclusions. Reporting the population, candidate rule, selected count, event denominator, and fixed-budget contrast makes these policy claims auditable.

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A Outcome, Model, and Cohort Details

Table 7: Baseline-oriented birth-certificate variables used by the frozen ranking instrument. Encounter-specific availability at months 1–3 requires validation in a prospective EHR source.

Feature	CDC source meaning
Maternal race	Maternal characteristic on birth record
Maternal height	Maternal characteristic on birth record
Pre-pregnancy BMI	Derived from pre-pregnancy measures
Pre-pregnancy weight	Weight before pregnancy
Previous live births now living	Previous live-born infants still living
Previous live births now dead	Previous live-born infants no longer living
Other pregnancy outcomes	Non-live-birth outcomes; may include current pregnancy
Pre-pregnancy diabetes	Diabetes diagnosed before this pregnancy
Pre-pregnancy hypertension	Hypertension diagnosed before this pregnancy

For CDC 2023 preterm records, early entry contains 2,601,060 records and 217,632 events; later entry contains 739,298 and 58,282; no care contains 76,520 and 17,387; unknown timing contains 63,504 and 9,700. Target-specific counts differ slightly because each outcome has its own missing-label exclusions. Source code 99 is normalized to missing before cohort construction, while source value 0 denotes no prenatal care.

B Complete Four-Cell Results

The point-estimate budgets at $q = 5\%$ are 130,053/174,019 for preterm, 129,864/173,829 for NICU, and 130,005/174,016 for low birth weight (early/all-record). At $q = 10\%$ they are 260,106/348,038, 259,727/347,658, and 260,009/348,032. These observed integer budgets remain fixed across empirical resamples.

Table 8: Complete symmetric-allocation results. All effects are percentage-point changes in population event capture; brackets give central 95% paired empirical-resampling ranges conditional on frozen scores and observed budgets.

Outcome	q	Joint Δ	ϕ_E	ϕ_B	s_B (%)	Interaction
Preterm birth	5%	3.37 [3.30, 3.43]	1.16 [1.09, 1.22]	2.21 [2.17, 2.25]	65.6 [64.3, 67.1]	0.33 [0.26, 0.41]
Preterm birth	10%	5.67 [5.58, 5.77]	2.15 [2.06, 2.24]	3.52 [3.48, 3.57]	62.1 [61.1, 63.3]	0.51 [0.44, 0.60]
NICU admission	5%	3.31 [3.26, 3.38]	1.19 [1.13, 1.25]	2.12 [2.09, 2.16]	64.1 [62.8, 65.4]	0.29 [0.23, 0.36]
NICU admission	10%	5.59 [5.51, 5.67]	2.11 [2.03, 2.20]	3.47 [3.44, 3.52]	62.2 [61.1, 63.3]	0.42 [0.34, 0.51]
Low birth weight	5%	3.96 [3.89, 4.06]	1.43 [1.34, 1.52]	2.53 [2.50, 2.57]	63.9 [62.3, 65.4]	0.40 [0.32, 0.49]
Low birth weight	10%	6.77 [6.67, 6.89]	2.60 [2.49, 2.72]	4.17 [4.12, 4.22]	61.6 [60.4, 62.7]	0.66 [0.55, 0.78]

Table 9: Auditable CDC 2023 four-cell point estimates. Each cell stacks population capture (percent) above selected true events in parentheses. Budgets stack early-sized above all-record-sized exact selected counts.

Outcome	q	Budgets b_e/b_a	$C_{00}: S_e, b_e$	$C_{01}: S_e, b_a$	$C_{10}: S_a, b_e$	$C_{11}: S_a, b_a$
Preterm birth	5%	130,053 174,019 260,106	9.05 (27,434)	11.10 (33,618)	10.04 (30,432)	12.42 (37,631)
Preterm birth	10%	348,038 129,864 173,829	14.63 (44,326)	17.89 (54,216)	16.52 (50,061)	20.30 (61,503)
NICU admission	5%	259,727 347,658	8.51 (25,801)	10.48 (31,795)	9.55 (28,967)	11.82 (35,854)
NICU admission	10%	347,658 174,016 260,009	13.93 (42,252)	17.20 (52,160)	15.84 (48,033)	19.52 (59,201)
Low birth weight	5%	348,032 174,016 260,009	9.53 (23,229)	11.86 (28,918)	10.76 (26,224)	13.49 (32,892)
Low birth weight	10%	348,032 174,016 260,009	16.06 (39,145)	19.90 (48,505)	18.33 (44,679)	22.83 (55,650)

The camera-ready replication uses 3,512,780 preterm records (307,612 events), 3,510,341 NICU records (310,650 events), and 3,514,422 low-birth-weight records (245,967 events) from CDC 2024. Models, preprocessing, and thresholds remain frozen from 2022. The corrected numeric MRACE31 representation matches model fitting; replay on CDC 2023 reproduces all 24 original four-cell point estimates exactly. Both full-precision and six-decimal threshold results, all 24 replication cells, and 3,000 paired-resample rows are archived.

Correction to the reviewed 2024 calculation. The original threshold-only runner loaded numeric maternal-race codes as strings, which the frozen numeric one-hot encoder mapped to unknown categories. At the nominal 10% cutoff, corrected full-precision selected counts are 372,390 / 373,434 / 358,394 for preterm / NICU / low birth weight, replacing 373,946 / 333,413 / 291,726. In particular, the earlier under-capacity results for NICU and low birth weight are superseded: their corrected worklists exceed the nominal budget. On the corrected scores, using six-decimal rather than full-precision thresholds changes these counts by only +1 / −8 / 0. The correction affects the supplementary 2024 calculation; the original CDC 2023 four-cell estimates are reproduced exactly.

C Synthetic Grid Details

Data-generating process. Let $g(x) = (1 + e^{-x})^{-1}$. For each evaluation record draw mutually independent $Z_i, \epsilon_i \sim \mathcal{N}(0, 1)$ and $U_i^Y, U_i^A \sim \text{Uniform}(0, 1)$. Define

$$Y_i = \mathbf{1}\{U_i^Y < g(\alpha_Y + Z_i)\}, \quad R_i = Z_i + \sigma\epsilon_i, \quad (8)$$

$$A_i = \mathbf{1}\{U_i^A < g(\alpha_A(a, \gamma) + \gamma Z_i)\}. \quad (9)$$

The full population contains all records and the available set is $\{i : A_i = 1\}$. The intercepts solve $\mathbb{E}[g(\alpha_Y + Z)] = 0.10$ and $\mathbb{E}[g(\alpha_A + \gamma Z)] = a$ for $Z \sim \mathcal{N}(0, 1)$. The outcome-risk coefficient is 1.0; $\alpha_Y = -2.5642215001$. Availability slopes $\gamma = -1, 0, 1$ encode higher risk being less available, independent of availability, or more available, respectively.

Table 10: Availability intercepts used in every synthetic condition. Both nonzero slopes have the same intercept by normal symmetry.

Target availability a	$\gamma = 0$	$\gamma = \pm 1$
0.5	0.0000000000	0.0000000000
0.7	0.8472978604	1.0184006520
0.9	2.1972245773	2.5642215001

The implementation uses 80-node Gauss–Hermite quadrature and bisection on $[-40, 40]$ with marginal-probability tolerance 10^{-13} and at most 200 iterations. Table 10 displays rounded values; the code solves them at runtime. The grid crosses $a \in \{0.5, 0.7, 0.9\}$, $\gamma \in \{-1, 0, 1\}$, $\sigma \in \{0.5, 1, 2\}$, and $q \in \{0.05, 0.10, 0.20\}$.

Repetitions and shared random draws. NumPy SeedSequence(2028).spawn(100) supplies independent repetition seeds to default_rng. Each repetition draws, in order, 100,000 evaluation risks, outcome uniforms, availability uniforms, and standard-normal score errors, followed by 50,000 independent threshold-selection risks and errors. The same base draws are reused across the entire grid: changing γ or a changes the availability rule, changing σ rescales the score error, and changing q changes the budget. Outcomes, scores, and masks remain shared between the four cells within a condition. Conditions are therefore paired within each repetition.

Selection and reported uncertainty. Budgets use Python’s round of q times the relevant pool size. Descending score order breaks exact ties by ascending original row position. For the transported-threshold policy, the independent threshold sample uses $R^{\text{cal}} = Z^{\text{cal}} + \sigma \epsilon^{\text{cal}}$; its round(q 50,000)-th highest score is applied with $\geq$. Every policy uses the same realized full-population event denominator within a repetition. Condition means and 2.5th/97.5th percentiles summarize 100 repetitions. These percentiles describe variation across generated populations, rather than a confidence interval for the Monte Carlo mean. The 18 sign reversals compare condition means; the nine negative ranges refer specifically to the all-record-budget eligibility contrast $C_{11} - C_{01}$. The 81 grid settings carry equal weight in the reported 22% reversal fraction, which describes this stress-test design rather than a prevalence of failure in clinical deployments.

The score-noise levels $\sigma = 0.5, 1, 2$ imply expected correlations between score and latent risk of approximately 0.894, 0.707, and 0.447. The largest absolute replicate-level interaction is 10.72 percentage points. Full condition summaries and replicate outputs are retained in the prepared release bundle.

D Audit Identities and Artifacts

Deterministic identity checks. For fixed S , scores, and deterministic tie breaking, $\text{Top}(S, b_e) \subseteq \text{Top}(S, b_a)$ whenever $b_e \leq b_a$. Increasing the budget adds exactly $b_a - b_e$ ranked positions and between zero and $b_a - b_e$ events. Both marginal capacity contrasts are therefore non-negative, as is their average ϕ_B . Substituting Equation 6 also gives $\phi_E + \phi_B = C_{11} - C_{00}$ exactly. These statements concern a shared score vector; retraining introduces an additional factor.

Artifacts. All manuscript figures and tables are generated from included CSV files. Verification scripts assert the six primary Shapley rows, 500 paired empirical resamples per target and budget, exact accounting identities, all 81 synthetic conditions, 18 mean sign reversals, and 9 conditions whose central replicate range is negative. The prepared release bundle contains aggregate inputs, figure and table generators, verification scripts, the locked synthetic configuration and runner, synthetic replicate outputs, and the corrected CDC 2024 replication aggregates. Local fitting and temporal-audit entypoints are provided in the repository. The package redistributes no individual-level Nataliy records. Use of the public-use files follows the NCHS Data

User Agreement [National Center for Health Statistics, 2025]; LightGBM is distributed under the MIT License [Microsoft Corporation and LightGBM Contributors, 2026]. Our code is released under the MIT License, separately from the CDC data terms and the paper’s license, at <https://github.com/vache4ogan/prenatal-risk-scoreability>.

Raw-data reconstruction. The CDC 2022 split is drawn before target-specific exclusions with `default_rng(2026)`: 20% enters validation and the remainder enters training, on which preprocessing is fitted. Unknown outcomes are excluded target-wise, and original row order breaks score ties. Thresholds are selected on the target-specific CDC 2022 all-record validation pool and applied with $\geq$. Empirical and synthetic seeds are 2027 and 2028. Section 3 and Appendices A–C specify the remaining cohort, model, and estimand details.

Compute and verification. The camera-ready checks used an Intel Core i5-8265U CPU (four cores, eight logical threads) with 8 GB installed RAM and no GPU. Frozen-model replay on CDC 2023 took 181 s; the full CDC 2024 audit with four worker threads and 500 paired resamples per outcome took 1,030 s. Repeating the complete 81-condition synthetic experiment took 81 s. Thus these three validation jobs required approximately 22 minutes of wall time in total, excluding data preparation and model fitting. The archived canonical analysis took 1,542 s. Original training-worker hardware and total exploratory compute were not retained. Verification covered frozen-model replay, aggregate arithmetic, and a fresh synthetic rerun; the full raw-file-to-model training pipeline was not rerun for the camera-ready. The repository distinguishes these routes and records environments, commands, and artifact hashes.

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